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checkpoint of chapter 1

1.1 — Original Question

Write a program that displays Welcome to Java, Welcome to Computer Science, and Programming is fun.

Solution — Java Code

```
public class Exercise1_1 {
    public static void main(String[] args) {
        System.out.println("Welcome to Java");
        System.out.println("Welcome to Computer Science");
        System.out.println("Programming is fun");
    }
}
```

Result / Output

```
Welcome to Java
Welcome to Computer Science
Programming is fun
```

1.2 — Original Question

Write a program that displays Welcome to Java five times.

Solution — Java Code

```
public class Exercise1_2 {
    public static void main(String[] args) {
        System.out.println("Welcome to Java");
        System.out.println("Welcome to Java");
        System.out.println("Welcome to Java");
        System.out.println("Welcome to Java");
        System.out.println("Welcome to Java");
    }
}
```

Result / Output

```
Welcome to Java
Welcome to Java
Welcome to Java
Welcome to Java
Welcome to Java
```

1.3 — Original Question

Write a program that displays the following pattern: the letters J, A, V, V, A in the shown arrangement.

Solution — Java Code

```
public class Exercise1_3 {
    public static void main(String[] args) {
        System.out.println("      J     A     V     V     A");
        System.out.println("      J    A A    V   V    A A");
        System.out.println("  J   J   AAAAA    V V    AAAAA");
        System.out.println("   J J   A     A    V     A     A");
    }
}
```

Result / Output

```
      J     A     V     V     A
      J    A A    V   V    A A
  J   J   AAAAA    V V    AAAAA
   J J   A     A    V     A     A
```

1.4 — Original Question

Write a program that displays a table of a, a^2, and a^3 for a = 1, 2, 3, 4.

Solution — Java Code

```
public class Exercise1_4 {
    public static void main(String[] args) {
        System.out.println("a   a^2   a^3");
        System.out.println("1    1     1");
        System.out.println("2    4     8");
        System.out.println("3    9     27");
        System.out.println("4    16    64");
    }
}
```

Result / Output

```
a  a^2  a^3
1   1    1
2   4    8
3   9    27
4   16   64
```

1.5 — Original Question

Write a program that displays the result of (9.5 × 4.5 − 2.5 × 3) / (45.5 − 3.5).

Solution — Java Code

```
public class Exercise1_5 {
    public static void main(String[] args) {
        double result =
                (9.5 * 4.5 - 2.5 * 3) / (45.5 - 3.5);
        System.out.println(result);
    }
}
```

Result / Output

```
0.8392857142857143
```

1.6 — Original Question

Write a program that displays the result of 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9.

Solution — Java Code

```
public class Exercise1_6 {
    public static void main(String[] args) {
        int result = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9;
        System.out.println(result);
    }
}
```

Result / Output

```
45
```

1.7 — Original Question

Write a program that approximates pi using the given series and displays two approximations.

Solution — Java Code

```
public class Exercise1_7 {
    public static void main(String[] args) {
        double pi1 = 4.0 * (
            1.0 - 1.0 / 3 + 1.0 / 5
            - 1.0 / 7 + 1.0 / 9 - 1.0 / 11
        );
        double pi2 = 4.0 * (
            1.0 - 1.0 / 3 + 1.0 / 5
            - 1.0 / 7 + 1.0 / 9 - 1.0 / 11
            + 1.0 / 13
        );
        System.out.println(pi1);
        System.out.println(pi2);
```

```
    }
}
```

Result / Output

```
2.9760461760461765
3.283738483738484
```

1.8 — Original Question

Write a program that calculates the area and perimeter of a circle with radius 5.5.

Solution — Java Code

```
public class Exercise1_8 {
    public static void main(String[] args) {
        double radius = 5.5;
        double perimeter = 2 * radius * Math.PI;
        double area = radius * radius * Math.PI;
        System.out.println("Perimeter = " + perimeter);
        System.out.println("Area = " + area);
    }
}
```

Result / Output

```
Perimeter = 34.55751918948772
Area = 95.03317777109125
```

1.9 — Original Question

Write a program that calculates the area and perimeter of a rectangle with width 4.5 and height 7.9.

Solution — Java Code

```
public class Exercise1_9 {
    public static void main(String[] args) {
        double width = 4.5;
        double height = 7.9;
        double area = width * height;
        double perimeter = 2 * (width + height);
        System.out.println("Area = " + area);
        System.out.println("Perimeter = " + perimeter);
    }
}
```

Result / Output

```
Area = 35.550000000000004
Perimeter = 24.8
```

1.10 — Original Question

Write a program that calculates the average speed in miles per hour for 14 kilometers in 45 minutes and 30 seconds.

Solution — Java Code

```
public class Exercise1_10 {
    public static void main(String[] args) {
        double kilometers = 14;
        double minutes = 45 + 30.0 / 60;
        double miles = kilometers / 1.6;
        double hours = minutes / 60;
        double speed = miles / hours;
        System.out.println(
            "Average speed = " + speed
            + " miles per hour"
        );
    }
}
```

Result / Output

```
Average speed = 11.666666666666668 miles per hour
```

1.11 — Original Question

Write a program that projects the U.S. population for each of the next five years.

Solution — Java Code

```
public class Exercise1_11 {
    public static void main(String[] args) {
        long population = 312032486;
        int secondsPerYear = 365 * 24 * 60 * 60;
        double births = secondsPerYear / 7.0;
        double deaths = secondsPerYear / 13.0;
        double immigrants = secondsPerYear / 45.0;
        for (int year = 1; year <= 5; year++) {
            population += (long)(births - deaths + immigrants);
            System.out.println(
                "Year " + year + ": " + population
            );
        }
    }
}
```

Result / Output

```
Year 1: 314812582
Year 2: 317592678
Year 3: 320372774
Year 4: 323152870
Year 5: 325932966
```

1.12 — Original Question

Write a program that calculates the average speed in kilometers per hour for 24 miles in 1 hour, 40 minutes, and 35 seconds.

Solution — Java Code

```
public class Exercise1_12 {
    public static void main(String[] args) {
        double miles = 24;
        double hours = 1;
        double minutes = 40;
        double seconds = 35;
        double timeInHours =
            hours + minutes / 60.0 + seconds / 3600.0;
        double kilometers = miles * 1.6;
        double speed = kilometers / timeInHours;
        System.out.println(
            "Average speed = " + speed
            + " kilometers per hour"
        );
    }
}
```

Result / Output

```
Average speed = 20.752614210336875 kilometers per hour
```

1.13 — Original Question

Solve the following 2 × 2 linear equations using Java: 3.4x + 50.2y = 44.5 and 2.1x + 0.55y = 5.9.

Solution — Java Code

```
public class Exercise1_13 {
    public static void main(String[] args) {
        double a = 3.4;
        double b = 50.2;
```

```
        double c = 2.1;
        double d = 0.55;
        double e = 44.5;
        double f = 5.9;

        double determinant = a * d - b * c;
        double x = (e * d - b * f) / determinant;
        double y = (a * f - e * c) / determinant;

        System.out.printf("x = %.4f%n", x);
        System.out.printf("y = %.4f%n", y);
    }
}
```

Result / Output

```
x = 2.6239
y = 0.7087
```